PERFECTNESS OF THE DERIVED PUSHFORWARD FOR A FAMILY OF BOUNDED COHERENT COMPLEXES WITH PROPER SUPPORT AND FINITE TOR DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- A bounded coherent complex on a Noetherian scheme pushes forward to a bounded coherent complex, and on Noetherian schemes bounded coherent means pseudo-coherent. invoked at proof lines 7001-7006. Reduces perfectness of Rf_*E to finite tor dimension.
- A pseudo-coherent complex is perfect exactly when it has finite tor dimension, detected by universally bounded cohomology after tensoring with all quasi-coherent sheaves. invoked at proof lines 7007-7011. Provides the criterion the rest of the proof verifies.
- The union of the supports of the cohomology sheaves is proper over the base and contained in a quasi-compact open, and the complex restricts to zero outside its support. invoked at proof lines 7013-7022. Allows replacing X by a quasi-compact open, making f quasi-compact and quasi-separated.
- The derived tensor of the pushforward with a quasi-coherent sheaf identifies with the pushforward of the tensor with the pulled back sheaf, and relative tor dimension bounds the cohomology of the inner tensor. invoked at proof lines 7025-7035. Concludes the universal boundedness of the cohomology.

Source reference. Original lemma 08E2: sources/stacks-perfect/pinned-original-source.tex, source lines 2852–2852. Cited at proof lines 7001-7003 for Rf_*E in D^b_Coh; the pseudo-coherence and tor dimension reduction are inside the counted span.

Source reference. Original lemma 08E8: sources/stacks-perfect/pinned-original-source.tex, source lines 2480–2480. Cited at proof lines 7004-7006 for pseudo-coherence of Rf_*E; the perfectness criterion is applied inside the counted span.

Source reference. Original lemma 08CQ: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 7005-7006 to reduce to tor dimension; the verification of tor dimension is inside the counted span.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08EA: sources/stacks-perfect/pinned-original-source.tex, source lines 2600–2600. Cited at proof lines 7008-7011 for the criterion; the verification through the base change rewriting is inside the counted span.

Source reference. Original lemma 0CYR: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 7014-7017 for T proper over S; the choice of the quasi-compact open X' is inside the counted span.

Source reference. Original lemma 01T7: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 7023-7024 for f quasi-separated after the restriction; the base change rewriting is inside the counted span.

Source reference. Original lemma 08EU: sources/stacks-perfect/pinned-original-source.tex, source lines 5269–5269. Cited at proof lines 7029-7032 for the first equality; the relative tor dimension bound concluding the proof is inside the counted span.

Source reference. Original lemma 08DE: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 7030-7033 for the second equality; the boundedness conclusion is inside the counted span.

Lemma 0.2. *Let S be a Noetherian scheme. Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $E \in D(\mathcal{O}_X)$ such that*

1. *$E \in D^b_{Coh}(\mathcal{O}_X)$,*
2. *the support of $H^i(E)$ is proper over S for all i, and*
3. *E has finite tor dimension as an object of $D(f^{-1}\mathcal{O}_S)$.*

Then Rf_*E is a perfect object of $D(\mathcal{O}_S)$.

Proof. By Lemma 08E2 we see that Rf_*E is an object of $D^b_{Coh}(\mathcal{O}_S)$. Hence Rf_*E is pseudo-coherent (Lemma 08E8). Hence it suffices to show that Rf_*E has finite tor dimension, see Cohomology, Lemma 08CQ. By Lemma 08EA it suffices to check that $Rf_*(E) \otimes^{\mathbf{L}}_{\mathcal{O}_S} \mathcal{F}$ has universally bounded cohomology for all quasi-coherent sheaves $\mathcal{F}$ on S. Bounded from above is clear as $Rf_*(E)$ is bounded from above. Let $T \subset X$ be the union of the supports of $H^i(E)$ for all i. Then T is proper over S by assumptions (1) and (2), see Cohomology of Schemes, Lemma 0CYR. In particular there exists a quasi-compact open $X' \subset X$ containing T. Setting $f' = f|_{X'}$ we have $Rf_*(E) = Rf'_*(E|_{X'})$ because E restricts to zero on $X \setminus T$. Thus we may replace X by X' and assume f is quasi-compact. Moreover, f is quasi-separated by Morphisms, Lemma 01T7. Now

$$Rf_*(E) \otimes^{\mathbf{L}}_{\mathcal{O}_S} \mathcal{F} = Rf_* \left(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} Lf^*\mathcal{F}\right) = Rf_* \left(E \otimes^{\mathbf{L}}_{f^{-1}\mathcal{O}_S} f^{-1}\mathcal{F}\right)$$

by Lemma 08EU and Cohomology, Lemma 08DE. By assumption (3) the complex $E \otimes^{\mathbf{L}}_{f^{-1}\mathcal{O}_S} f^{-1}\mathcal{F}$ has cohomology sheaves in a given finite range, say $[a, b]$. Then Rf_* of it has cohomology in the range $[a, \infty)$ and we win. □